

$$RR = \frac{y_0}{y_c} = (1 - y_0) \times e^{-\Delta PM \cdot \beta} + y_0$$

Assuming that:

$$e^{-\Delta PM \cdot \beta} \cong (1 - y_0) \times e^{-\Delta PM \cdot \beta} + y_0$$

$$\Rightarrow RR \cong e^{-\Delta PM \cdot \beta}$$

It is then possible to calculate the underlying coefficient:

$$\frac{\ln(RR)}{-\Delta PM} \cong \beta$$

$$\Rightarrow \beta_3 = \frac{\ln(1.81)}{45} = 0.01319$$

Since this coefficient estimate is based on the assumption that

$$e^{-\Delta PM \cdot \beta} \cong (1 - y_0) \times e^{-\Delta PM \cdot \beta} + y_0 ,$$

it should be used in a C-R function that maintains this assumption. In effect, it should be applied to a log-linear C-R function:

$$\Delta y = \left[y_0 \times \left(e^{\beta \cdot \Delta PM} - 1 \right) \right]$$

Using the formula for the change in the incidence rate and assuming a 10 µg/m³ decline in PM_{2.5}, it is shown that this results in changes within the bounds suggested by the two estimates based on using the estimated odds ratios:

$$\Delta y_1 = \frac{.0717}{(1 - 0.0717) \times e^{10 \times 0.01462} + 0.0717} - 0.0717 = -0.00914$$

$$\Delta y_2 = \frac{.0717}{(1 - 0.0717) \times e^{10 \times 0.01394} + 0.0717} - 0.0717 = -0.00874$$

$$\Delta y_3 = 0.0717 \times \left(e^{-10 \times 0.01319} - 1 \right) = -0.00886$$

In this instance, it seems that simply using the relative risk to estimate the underlying coefficient results in a good approximation of the change in incidence. Since it is unclear which of the two other coefficients (β_1 or β_2) should be used - as the published work was not explicit - the coefficient based on the relative risk and the log-linear functional form is a reasonable approach.

C.6 Cox Proportional Hazards Model

Use of a Cox proportional hazards model in an epidemiological study results in a C-R function that is log-linear in form. It is often used to model survival times, and as a result, this discussion focuses on mortality impacts.

The Cox proportional hazards model is based on a hazard function, defined as the probability that an individual dies at time t , conditional on having survived up to time t (Collet, 1994, p. 10). More formally, the hazard function equals the probability density function for the risk of dying divided by one minus the cumulative probability density function:

$$h(X, t) = \frac{f(X, t)}{1 - F(X, t)}$$

The proportional hazards model takes the form:

$$h(X, t) = h_0(t) e^{X \cdot \beta},$$

where X is a vector of explanatory variables, β is a vector of coefficients, and $h_0(t)$ is the so-called “baseline hazard” rate. This terminology differs from that used in most of this discussion: this “baseline hazard” is the risk when all of the covariates (X) are set to zero; this is *not* the risk in the baseline scenario.

The Cox proportional hazards model is sometimes termed a “semi-parametric” model, because the baseline hazard rate is calculated using a non-parametric method, while the impact of explanatory variables is parameterized. Collet (1994) details the estimation of Cox proportional hazards models; in particular, see Collet’s discussion (pp. 95-97) of nonparametric estimation of the baseline hazard.

Taking the ratio of the hazard functions for the baseline and control scenarios gives the relative risk:

$$RR = \frac{h(X_0, t)}{h(X_c, t)} = \frac{h_0(t) e^{X_0 \cdot \beta}}{h_0(t) e^{X_c \cdot \beta}} = e^{\Delta PM \cdot \beta},$$

where it is assumed that the only difference between the baseline and control is the level of PM pollution.

The relative risk is often presented rather than the coefficient β , so it is necessary to estimate β in order to develop the functional relationship between ΔPM and Δy , as described previously for log-linear C-R functions.

Appendix D. Health Incidence & Prevalence Data in U.S. Setup

Health impact functions developed from log-linear or logistic models estimate the percent change in an adverse health effect associated with a given pollutant change. In order to estimate the absolute change in incidence using these functions, we need the baseline incidence rate of the adverse health effect. And for certain health effects, such as asthma exacerbation, we need a prevalence rate, which estimates the percentage of the general population with a given ailment like asthma. This appendix describes the data used to estimate baseline incidence and prevalence rates for the health effects considered in this analysis.

D.1 Mortality

This section describes how we developed county mortality rates for the years 2015 through 2050 to use in BenMAP. First, we describe the source of 2012-2014 baseline mortality data and how we calculated county-level mortality rates. We then describe how we used national-level Census mortality rate projections to develop county-level mortality rate projections for years 2015-2060.

D.1.1 Mortality Data for 2012-2014

We obtained county-level mortality and population data from 2012-2014 for 11 causes for the contiguous United States by downloading the data from the Centers for Disease Control (CDC) WONDER database (<http://wonder.cdc.gov>).

Since the detailed mortality data obtained from CDC do not include population, we combined them with U.S. Census Bureau population estimates exported from BenMAP. We then generated age-, cause-, and county-specific mortality rates using the following formula:

$$R_{i,j,k} = \frac{D_{i,j,k}(2012) + D_{i,j,k}(2013) + D_{i,j,k}(2014)}{P_{i,k}(2012) + P_{i,k}(2013) + P_{i,k}(2014)}$$

where $R_{i,j,k}$ is the mortality rate for age group i , cause j , and county k ; D is the death count; and P is the population.

For county-age group cells with fewer than 10 deaths, CDC WONDER suppresses the exact death count. For these observations, a mortality rate cannot be calculated. For each combination of age group and mortality cause, we used the following procedure to deal with suppressed counts.

For each combination of state, age group and mortality cause, we grouped counties with unsuppressed mortality figures and summed their reported death counts. We then subtracted these unsuppressed deaths from the state-level age- and cause-specific death count, which includes suppressed deaths. We divided the resulting state-wide death count in suppressed counties by the age-specific populations in those counties. This calculation resulted in an age- and cause- specific average mortality rate for suppressed counties;